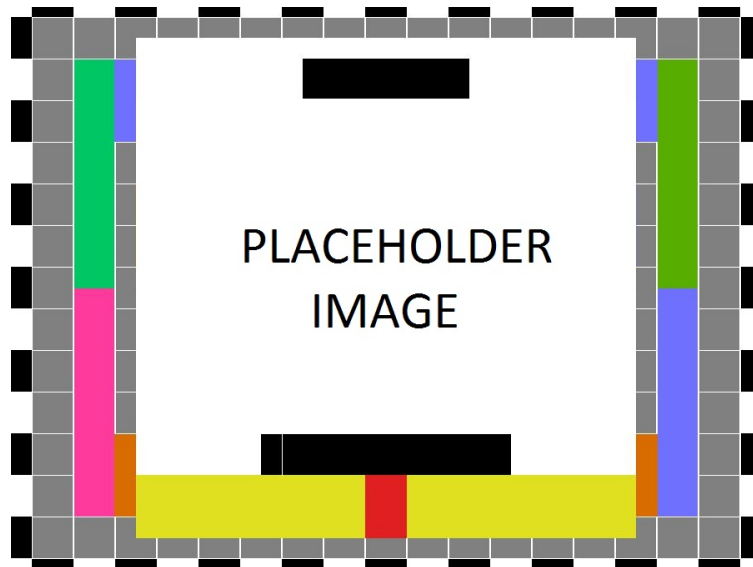


**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**DETAILED DESIGN SPECIFICATION
CSE 4317: SENIOR DESIGN II
SPRING 2027**



**TEAM FULLSTACK
CIVICPULSE COMMUNITY SERVICE-REQUEST
DASHBOARD**

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REVISION HISTORY

Revision	Date	Author(s)	Description
0.1	01.20.2027	JVN	document creation
1.0	02.05.2027	AT, GH, CB	official release

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1 INTRODUCTION

Implementation-level design of CivicPulse Community Service-Request Dashboard as an IEEE 1016-2009 [1] Software Design Description, building on the SRS and ADS (tiers X/Y/Z).

2 DESIGN STAKEHOLDERS & CONCERNS

Implementers, integrators, testers, city IT (maintainers); concerns of composition, data, interfaces, state, algorithms, and resources.

3 DESIGN VIEWPOINTS

Context, Composition, Information, Interface, Interaction & State, Algorithm, Resource per IEEE 1016; omitted where not applicable.

4 SYSTEM OVERVIEW

Frontend: React/TypeScript SPAs served by a CDN/reverse proxy. Backend: containerized Node/TypeScript (NestJS) API. Data: managed PostgreSQL + PostGIS and object storage.

5 X LAYER SUBSYSTEMS — PRESENTATION

5.1 LAYER RESOURCES

Deps: React, TypeScript, a map library; **Language:** TypeScript.

5.2 X.RESIDENTPORTAL & X.STAFFDASHBOARD

Composition: SPA route modules (submit, track, dashboard, request detail). **Interface (#01):** calls the REST API; renders map via EI-02. **State:** client form/query state; auth token held in memory for staff. **Interaction:** optimistic status updates with server confirmation. **Rationale:** SPA split enables responsive/accessible UI (US-01) over the stateless API (AD-01).

6 Y LAYER SUBSYSTEMS — APPLICATION

6.1 LAYER RESOURCES

Deps: NestJS, an OIDC client, an ORM, an SMTP client; **Language:** TypeScript.

6.2 Y.APIGATEWAY

Interface (#02): REST; validates OIDC access tokens (EI-01) and enforces RBAC middleware (SE-01). **Algorithm:** per-route role guard; deny by default.

6.3 Y.REQUESTSERVICE

Information: Request{id, category, description, geom(Point), photo_ref, status, priority, assignee, history[]}. **Interface:** CRUD + triage/assign endpoints. **State:** status machine NEW→TRIAGED→ASSIGNED→IN_PROGRESS →RESOLVED→CLOSED. **Algorithm (FR-05):** on create, ST_DWithin PostGIS query for same-category requests within radius/time; flag matches. **Rationale:** PostGIS does the spatial work natively.

6.4 Y.NOTIFICATIONSERVICE

On status change, render a template and send via SMTP (FR-04); failures retried and logged (MS-01).

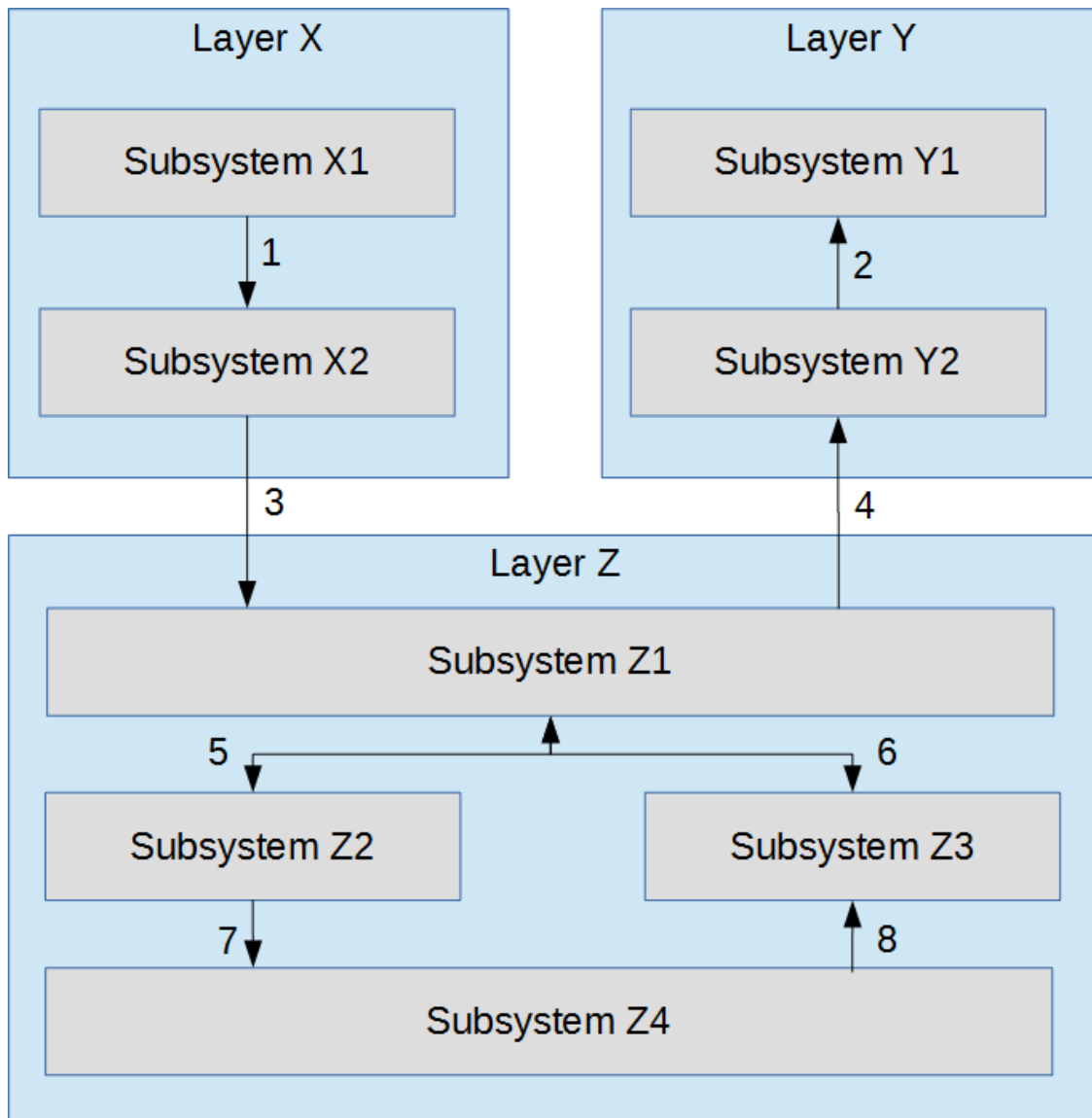


Figure 1: Three-tier deployment carried forward from the ADS

7 Z LAYER SUBSYSTEMS — DATA

7.1 LAYER RESOURCES

Tech: PostgreSQL + PostGIS; object storage; **Language:** SQL/migrations.

7.2 Z.DATASTORE

Information: normalized tables `requests`, `users`, `assignments`, `status_history`, `audit`; spatial index on `geom`. **Interface:** ORM + migrations.

7.3 Z.INTEGRATIONS

Adapters for geocoding (EI-02) and SMTP behind interfaces so providers can be swapped without touching services.

8 REQUIREMENTS TRACEABILITY

Req	Requirement (abbrev.)	Component	Iface
FR-01/02	Submit/track	X.ResidentPortal, Y.RequestService	#01
FR-03	Triage/assign	Y.RequestService	#02
FR-04	Status/notify	Y.NotificationService	—
FR-05	Duplicates	Y.RequestService, Z.Datastore	—
SE-01	RBAC+TLS	Y.APIGateway	#02

9 APPENDIX A

Database schema, OpenAPI specification, and infrastructure-as-code are attached to the closeout package.

REFERENCES

- [1] IEEE, “IEEE Std 1016-2009 — IEEE Standard for Information Technology — Systems Design — Software Design Descriptions,” Institute of Electrical and Electronics Engineers, New York, NY, USA, Tech. Rep., 2009.